

FallGuard: Individual Contributions and Reflections

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Abstract—This document consolidates the five members’ contribution statements and team task records for FallGuard, an edge-cloud fall-detection project using quality-aware bidirectional temporal modeling. It records individual work, collaboration, corresponding project deliverables, and learning outcomes. The contributions cover model development and deployment, data preparation, training, cloud delivery, interface development, testing, literature review, and project communication.

I. PROJECT AND TEAM ORGANIZATION

The final system extracts poses on a Raspberry Pi, classifies one-second skeleton windows with MaskedBiMamba, stores records locally, and synchronizes them to a persistent cloud archive. The descriptions below use the final system terminology; repository paths identify corresponding deliverables.

II. YAORONG HUANG

A. Model development and system integration

I served as group leader and carried out most of the project’s code implementation and integration. My work covered temporal modeling, pose preprocessing, training and evaluation tools, model export, and the connection between edge and cloud services. I implemented MaskedBiMamba by combining established Mamba operators with temporal attention, two temporal directions, frame masking, and quality-weighted pooling. I organized the baseline and component comparisons so that the model’s precision and tolerance to missing frames could be assessed alongside its clean-data performance.

I integrated USB-camera and video acquisition, Hailo-accelerated pose extraction with a CPU fallback, and the two-input ONNX temporal classifier on the Raspberry Pi. I implemented local alert-state transitions and corrected the pose-preprocessing order, updating the experiment archive and report to reflect that correction.

B. Workspaces, manuscript, and presentation

I completed the final local and cloud workspaces, with live preview, current results, historical charts, filters, pagination, CSV export, and diagnostics. I added freshness checks so that the displayed result follows the current input and service state. Deployment verification included preserving existing databases, checking running services and source hashes, and retaining rollback copies.

I took primary responsibility for the LaTeX report, including the method notation, experimental protocol, results, system implementation, and final editing. I maintained consistency between the manuscript, figures, result summaries, technical guides, and deployed software. I developed the twelve-slide

GitHub Pages presentation and its interactive charts, bilingual notes, presenter tools, and rehearsal timer. This web implementation complemented the group’s slide content and module-specific presentation preparation.

C. Problem and solution

The deployment had to keep inference and records available when cloud connectivity was lost. I separated local inference and SQLite storage from an independent outbox, then checked retry and recovery; the recorded delivery tests retained record identifiers and capture timestamps after reconnection.

D. Collaboration, deliverables, and reflection

I coordinated the model, data, delivery, and manuscript interfaces across the team.

Corresponding deliverables: `code/src/fallbench/`, `code/scripts/export_mamba_onnx.py`, `deployment/`, `artifacts/models/`, `manuscripts/final-report/`, and `docs/`.

The main lesson for me was to develop the model and its operating system together. Local persistence, recoverable delivery, clear status displays, and consistent documentation made the classifier useful as an IoT prototype. Maintaining one traceable chain from configuration to result table also made final integration and presentation more efficient.

III. JIAHAO ZHAO

A. Cloud interfaces and record delivery

My responsibilities covered the cloud interface, system testing, and project records. I worked on the interface through which classified edge records become a durable cloud history, including the record structure, storage behavior, and API access used by the detection interface. In the final system, temporal classification runs on the Pi and the cloud receives already classified records. The cloud-side workflow preserves the original observation time and acknowledges an identifier after the record has been stored, supporting safe retry after an interrupted transfer.

I contributed to the archive functions used for historical inspection: querying stored windows and events, filtering, pagination, and exporting records. I coordinated the device, timestamp, prediction, and alert fields with the edge implementation and the interface so that the recorded history and the displayed history describe the same data.

B. Testing and project coordination

I contributed system and regression tests for the record path, including malformed input handling, duplicate delivery, acknowledgement loss, restart persistence, and recovery after disconnection. The test evidence also covers archive access and the behavior of the actual service launchers. These checks supported the final demonstration by verifying the interaction between components, rather than assessing the classifier in isolation.

I maintained the consolidated meeting records, recording decisions, task owners, deadlines, and module handovers. Eight meetings between 9 September and 3 October documented the team’s progress. Meeting note-taking rotated among members, and I organized these records into the final set. I also collected service instructions, environment information, and test outputs as part of the reproducibility materials.

C. Problem and solution

A cloud write can succeed even when its acknowledgement is lost, leaving the edge to retry a record already stored. I worked with edge integration on commit-before-acknowledgement and UUID-based deduplication, and contributed retry and restart tests that checked for duplicate rows and preserved capture times.

D. Collaboration, deliverables, and reflection

I coordinated the record schema with Yaorong Huang and the displayed fields with Chengyi Xu, supplying system details for the report review.

Corresponding deliverables: `deployment/cloud/`, `webui/server.py`, `webui/cloud_workspace.py`, `tests/`, `results/deployment/`, and the group’s meeting records.

This work strengthened my understanding of durable delivery and service testing. Keeping storage and acknowledgement in the correct order, and exercising restart and retry behavior, turned recovery into a behavior we could demonstrate. Recording module handovers also helped the team resolve interface changes efficiently.

IV. CHENGYI XU

A. Training and experiment management

My responsibilities covered the training workflow and detection interface. I contributed the workflow that reads prepared skeleton sequences, forms training batches, addresses class imbalance, and manages checkpoints and metric logs. I worked with Yaorong Huang on the connection between the model definition and its training configuration, and with Jiayun Song on the input sequence format and grouped evaluation partitions.

I contributed training augmentation and random-seed management so that the comparison runs followed recorded settings. The final protocol trains for a fixed number of epochs, selects checkpoints by validation F1, and evaluates the same folds and seeds across models. The loss configuration, run commands, checkpoints, and metric outputs form the record of each experiment. I also supported the component comparisons

and the organization of their results into the report’s tables and figures.

B. Detection interface and demonstration

I contributed to the detection interface displaying device health, current predictions, and stored alerts, and to connecting those views to the cloud API. I worked on the initial interface panels and demonstration workflow. Yaorong Huang subsequently integrated and completed the final three-page local and cloud workspaces. This division connected my interface preparation with the final deployment and presentation.

I coordinated interface fields with Jiahao Zhao and checked that the displayed data matched stored records. I supported the demonstration material and the explanation of the detection workflow, including how an operator moves from current status to historical inspection. These activities made the project accessible to viewers who were not involved in model development.

C. Problem and solution

Fall windows were much rarer than daily-activity windows, creating a risk that training would favor the majority class. I used class-aware weighted focal-loss settings and recorded masking augmentation and random seeds in the shared workflow, so the comparison runs followed the same class-balancing and run protocol.

D. Collaboration, deliverables, and reflection

I coordinated data inputs with Jiayun Song, model settings with Yaorong Huang, and interface fields with Jiahao Zhao.

Corresponding deliverables: `code/src/fallbench/engine.py`, `code/src/fallbench/losses.py`, `code/src/fallbench/reproducibility.py`, `code/scripts/train.py`, `code/configs/`, and the detection-interface and demonstration materials.

I learned to treat configuration and logging as part of the experiment, rather than as supporting paperwork. Shared inputs, consistent run settings, and saved outputs made model comparisons easier to interpret. Interface work also showed me the value of displaying status, time, and history together when explaining a running system.

V. ZHONGLIN CHANG

A. Problem definition and literature review

I took responsibility for the problem definition, algorithm survey, and references. I framed the task around distinguishing a fall transition from daily activities with similar individual poses, such as sitting, bending, or kneeling. This motivated the use of temporal skeleton sequences and the explicit consideration of incomplete pose observations. I reviewed wearable, environmental, and camera-based approaches before focusing the comparison on pose-based temporal modeling and edge deployment.

I organized the candidate methods and their uses in the project. The review included TCNTE as a temporal baseline, masked temporal modeling as a motivation for frame masking, and confidence-aware skeleton approaches as context for the

quality input. I helped articulate MaskedBiMamba’s contribution as the combination of temporal context and pose quality, together with matched evaluation and practical edge–cloud integration.

B. References and report review

I maintained the shared reference list and checked the sources used in the report. The literature records connect the cited methods and datasets to their source documents and distinguish available implementation resources from papers used as methodological background. I contributed the problem-definition and related-work material and reviewed the alignment between the report’s claims and the team’s recorded results.

I coordinated the method, dataset, and system descriptions with the team to keep the introduction, method, and results consistent.

C. Problem and solution

Posture or motion thresholds alone can confuse rapid sitting with a fall, so selecting a model required a clear account of the temporal information it uses. I compared pose-based temporal candidates under the project’s camera and compute constraints, checked source papers and implementation resources, and wrote the selection rationale used in the problem definition and related work.

D. Deliverables and reflection

Corresponding deliverables: `manuscripts/final-report/texfiles/2.Introduction.tex`, `manuscripts/final-report/references.bib`, `manuscripts/proposal/REFERENCE_VERIFICATION.md`, and the candidate-method and reference notes.

I learned that a useful literature review must connect a research idea to an implementable comparison. Checking datasets, code availability, and the precise purpose of each citation helped the team choose practical baselines and explain its design clearly. Reviewing the report against the result records also improved the consistency of the final argument.

VI. JIAYUN SONG

A. Data preparation and evaluation inputs

My responsibilities covered dataset preparation, edge testing, and presentation materials. I organized dataset access information, labels, usage terms, and the prepared skeleton sequences used by the training workflow. The final primary evaluation uses CAUCAFall and GMDCSA-24, while Le2i provides a separate baseline comparison and an external test. I coordinated the input sequence format and the evaluation partitions with Chengyi Xu and Yaorong Huang.

I contributed to label mapping, subject-grouped partition preparation, and the conversion of videos into reusable pose sequences. Keeping a subject’s recordings within one partition supports a fair cross-subject comparison, and preparing the sequence once lets training reuse the same representation. The final materials publish the recovered test assignment for the 260 primary videos together with its provenance information.

B. Edge testing and presentation preparation

I contributed edge measurements and the organization of robustness and cross-dataset results. The paired hardware measurements compare Hailo and CPU pose processing at the same input resolution while retaining the same local classifier. I supported the collection and presentation of throughput, processing latency, memory, and temperature evidence, as well as the preparation of the synthetic edge-evaluation sample.

I prepared presentation content, figures, and speaker notes, helped maintain a consistent visual style, and supported rehearsal. Yaorong Huang implemented the final interactive web presentation; my work focused on communicating the datasets, system workflow, and evaluation results within that shared presentation. I also supported the preparation of demonstration and non-fall example material described in my individual work statement.

C. Problem and solution

The CPU pose path provided limited throughput for the intended edge workflow. I helped measure Hailo and CPU processing at the same input resolution using a paired replay; the archived full-pipeline results were 20.56 versus 1.73 frames/s, providing evidence for the accelerated deployment configuration.

D. Collaboration, deliverables, and reflection

I coordinated training inputs with Chengyi Xu and edge measurements with Yaorong Huang, and supplied timestamps and citations for the report.

Corresponding deliverables: `datasets/DATASET_PROVENANCE_AND_USAGE.md`, `datasets/annotation_sources.json`, `datasets/primary_test_folds.csv`, `code/scripts/make_splits.py`, `code/scripts/build_windows.py`, `results/edge/`, `results/robustness/`, and the presentation content and notes.

I learned to inspect data preparation and partitions before interpreting model scores. Clear dataset roles, recorded transformations, and consistent measurements made the comparisons easier to explain. Preparing the presentation also helped me turn technical results into a coherent account of how the data, model, and deployed system work together.